

# Yalun Wu

✉ aaaronwww@outlook.com

☎ (+86) 189-4058-1177 (+65) 9812-5849

📍 National University of Singapore, Singapore



## 🎓 Education

**Beihang University (BUAA)**

*B.Eng. in Cyberspace Security*

Sep 2021 – Jul 2025

GPA: 89/100

**National University of Singapore (NUS)**

*Master of Computing, NEXt++ Lab*

Aug 2025 – Dec 2026, *expected*

**Coursework:** Data Structures, Machine Learning, Deep Learning, RL

**Tests:** IELTS 6.5; GRE 322, V152 / Q170

## 🔗 Research Interests

Trustworthy LLM systems and agent architectures: multi-step RAG, self-evolution, memory governance, runtime verification, and evaluation under strict constraints. Experience spans problem formulation, benchmark design, implementation, deployment, and operational validation.

## 🔧 Technical Skills

- **Agent systems:** Workflows, harnesses, task decomposition, multi-agent collaboration, tool use, structured memory, and verifiable execution.
- **RAG:** Multi-step multimodal retrieval and long-context compression; optimizing recall, evidence retention, reasoning stability, latency, and cost.
- **Trustworthy evaluation:** Benchmarks, failure taxonomies, evidence-based scoring, model comparisons, ablations, and error attribution; reproducible tests with structured outputs, schema constraints, runtime checks, and physical/safety rules.
- **Multimodal models:** VLMs, medical imaging, embodied time series, visual RAG, and business search under structural and safety constraints.
- **Engineering:** Python, PyTorch, Linux, Git, LaTeX; deployment, data pipelines, batch experiments, visualization, scientific writing, and teamwork.

## 💼 Internship Experience

### ⚙️ Meituan

LLM Algorithm Intern

May 2026 – Present

Multi-step Visual RAG and Business Intelligence

- Built three-route hybrid retrieval and multi-step visual RAG for production knowledge-base QA; optimized evidence selection, cross-step memory, and visual inputs, raising Recall@3 from 51.9% to over 79%.
- Led 85+ controlled experiments on retrieval, inference settings, accumulated context, visual quality, and agent architectures; analyzed retrieval success, evidence use, and cost to develop reusable configuration and tuning procedures.
- Raised strict internal accuracy from 70.8% to 81.5%, up 10.7 pp; cut P50 inference latency from 164 s to 21 s.
- Extended to multimodal query rewriting and text-only QA, gaining 7/4 pp on business benchmarks; deployed query rewriting to production.

### ⚙️ Huazhu Group

Enterprise AI Agent Intern

Oct 2025 – Jan 2026

Food-service Supply Chain and AI Infrastructure

- Mapped centralized procurement, bulk purchasing, replenishment, and menu-planning workflows, rules, and data interfaces for supply-chain AI.
- Built multi-agent modules for procurement, product/supplier extraction, replenishment, and menu planning, linking data to recommendations.
- Consolidated store needs, prices, product constraints, and supplier policies through feedback and prototypes; explored structured skill generation and validation, informing LFUSE's scenarios and business rules.

### 🏢 Haitun Zhisheng Medical Technology Co., Ltd.

LLM Evaluation Algorithm Intern

Apr 2025 – Sep 2025

Vision-Language Model Evaluation

- Led ultrasound VLM deployment and benchmarking; ran experiments, verified results, and standardized outputs for model comparisons.
- Contributed to U2-BENCH and DOLPHIN through reasoning-mode comparisons, cross-anatomy generalization analysis, and ablations.
- Designed prompts, structured data, and analyzed outputs; wrote experiment and failure reports supporting paper figures and conclusions.

### ⚙️ Beijing Feixiong Technology Co., Ltd.

Intern

Jul 2024 – Sep 2024

UAV Research and Development

- Tested UAV communication security; supported troubleshooting, fault localization, and functional verification.

## 🔬 Research and Projects

*Agent Systems, RAG, and Business Intelligence*

### 🔍 TAEC: Trajectory-Aware Evidence Coordination for Multi-Step Visual RAG

First author

ICLR 2027, under review

- Proposed training-free trajectory-level evidence coordination for multi-step visual RAG. Shared unmet answer requirements align evidence admission, memory rendering, and visual detail allocation, reducing cross-step interference.
- Designed joint coverage and redundancy filtering, adaptive memory rendering, and requirement-aware visual budgeting to retain useful observations, compress resolved information, and prune failed branches.
- Achieved **66.8%** mean accuracy across 12 settings on 3 benchmarks and 4 backbones, **4.5 pp** above DAG; same-backbone gains on ViDoSeek, SlideVQA, and MMLongBench-Doc were **7.4/6.9/5.6 pp**.

### ⚙️ EvolveMAS: A Unified Self-Evolution Framework for Decentralized Multi-Agent Systems

First author

ICLR 2027, under review

- Proposed a self-evolution framework for decentralized multi-agent systems, unifying within-episode adaptation and cross-episode knowledge evolution across cooperative, mixed-motive, and adversarial settings.
- Paired each agent with a MemoryAgent managing episodic, semantic, and working memory outside its reasoning context; Execute–Evaluate–Evolve distills reusable strategies and partner profiles without updating model parameters.
- Achieved the highest normalized mean score of **77.2** across 5 benchmarks and 6 configurations, **7.7 points** above Self-Memory; factorial ablations isolated the contributions of memory governance and cross-episode knowledge reuse.

## ⚙️ How Much Specification Does External Supervision of an Agent Need?

Corresponding author

ICLR 2027, under review

- Studied executable task conditions for runtime supervision and false-completion detection; held models, tasks, and step budgets fixed to compare how condition coverage and composition affect completion and false acceptance.
- Compared within-task condition swaps, cross-model transfer, and feedback versus enforcement to assess failure-targeted supervision and distinguish monitor access to conditions from feedback given to agents.
- At 50% coverage, selecting conditions from another model's stable failure records improved strict task pass rates by **18.5/19.3 pp** over random selection for two models, while reducing false acceptance of incomplete tasks.

## 🗄️ LHM: Long-Horizon Memory Governance for LLM Agents

Co-first author

AAAI 2027, under review

- Addressed fact conflicts and memory drift through versioned revision, risk gating, constrained decoding, and governed forgetting.
- Versioning and immutable audit logs raised conflicting-fact update accuracy from 40% to 100%; NLI-based contradiction control improved cross-turn consistency by up to 13 pp in long-horizon and multi-session tests.

## 🔍 LFUSE: Learned Evidence Fusion for Noisy Spoken and Image-Grounded Product Search

Co-first author

ICASSP 2027, under review

- Helped design tasks, workflows, and agent prototypes for noisy speech/image search in Huazhu's procurement, replenishment, and menu planning.
- Turned user requirements, product constraints, supplier policies, and operational feedback into search tasks and evaluation protocols.

## 🗃️ SchemaBridge: A Unified Schema Interface and Corpus for Cross-Task Semantic Parsing

First author

NAACL 2027, planned submission

- Led unified semantic parsing and writing, mapping NER, RE, EE, and SLU to hierarchical category-args schemas for devices and business assistants.
- Unified 61 datasets and 2.75M multilingual examples; analyzed Intent Engine's Parallel Match, latency, and training-deployment schema mismatches.

## *Trustworthy Evaluation in Embodied and Safety-Constrained Settings*

### ✈️ FLY-EVAL++: An Evidence-Driven Evaluation Protocol for Safety-Constrained Flight Prediction

First author

COLM 2026

- Proposed a two-stage protocol for flight prediction under physical and safety constraints: deterministic evidence-atom checks of output structure, physical feasibility, and FAA rules, followed by interpretable rubric aggregation without LLM-based scoring.
- Built tasks with 34 sensor channels, multi-stage attitude-trajectory prediction, and 20 FAA operational constraints; extended evaluation to multi-step rollouts to analyze accuracy degradation and safety risks as temporal complexity increases.
- Screened 66 LLMs and fully evaluated 21, revealing safety-score gaps exceeding 28 points among similarly accurate models. Identified failures in protocol compliance, field validity, physical consistency, safety coverage, and prediction accuracy.
- Demonstrated why predictive accuracy alone misses operational reliability; established a reusable process for deriving metrics from domain rules, testing safety boundaries, and analyzing failures in other embodied applications.

### ✈️ PilotBench: A Benchmark for General Aviation Agents with Safety Constraints

First author

IJCNN 2026

- Built joint trajectory-attitude prediction tasks from 708 real flights, including preprocessing, model evaluation, and results analysis.
- Evaluated 41 models and conventional predictors; designed PILOT-SCORE for regression accuracy, instruction following, and safety compliance, leading experiments, verification, and comparisons.
- Analyzed accuracy-controllability trade-offs and dynamic failures, laying foundations for FLY-EVAL++ evaluation and FTAP modeling.

### ✈️ Safety-Constrained Temporal Flight Modeling with Foundation Models

Co-first author

Nature Machine Intelligence, planned submission

- Reconstruct FTAP time series and integrate public aviation data for a high-fidelity, physics-constrained long-horizon flight prediction benchmark.
- Distinguish feasible predictions from plausible but unsafe outputs; develop safety-constrained training objectives, ablations, and validation.

## *Domain-Specific Large Models and Multimodal Infrastructure*

### 👤 U2-BENCH: Benchmarking Large Vision-Language Models on Ultrasound Understanding

Core contributor; co-author

ICLR 2026

- Helped build an ultrasound benchmark with 7,241 images, diagnostic texts, and 8 tasks spanning classification, detection, regression, and generation.
- Evaluated 20 general-purpose and medical VLMs, verifying results and standardizing outputs for comparability and reproducibility.
- Analyzed performance and failures; prepared visualizations, comparisons, and key figures to explain differences in ultrasound understanding.

### ♥️ H2HTalk: Evaluating Large Language Models as Emotional Companion

Core contributor; second author

NLPCC 2025 Oral Paper

- Helped build an emotional-support and persona-consistency benchmark with scenario design, evaluation rules, and automated scoring; processed data and tested 49 models across 4,800+ multi-turn dialogue scenarios.
- Used ROUGE, BERTScore, and contextualized persona comparisons to analyze memory retrieval, emotional feedback, and implicit needs, revealing limits in long-term memory and user understanding.
- Studied safety and emotion regulation with attachment-theory-based SAP; removing it raised violation rates by an order of magnitude.

### 👤 DOLPHIN: A Multimodal Large Language Model for Ultrasound Understanding

Core contributor; listed immediately after co-first authors

CVPR 2027, planned submission

- Evaluated post-training, instruction tuning, and UARPO reinforcement learning; compared reasoning and standard modes across 16 anatomical regions, analyzing measurement errors and missed detections.
- U2-Score: 0.5835; reasoning versus standard mode raised diagnosis/detection accuracy by 2.4%/16% and reduced measurement RMSE by 10.6%.

### 🗄️ NeuroMark: Multi-Modal Evaluation for Neuropsychiatry with Joint Trustworthiness Metrics

Co-first author

Nature Communications, planned submission

- Help define medical-text reasoning and brain-image understanding tracks for neuropsychiatric evaluation and manuscript planning, including task scope, model coverage, and trustworthiness.
- Lead BrainVLM: integrate multi-source brain MRI; evaluate 16 VLMs on 6 tasks with unified inference, parsing, scoring, and cross-task analysis.